GEMINI API INTEGRATION DIAGNOSTIC REPORT

AI Research and Knowledge Workspace | Final Diagnostic Audit & Pipeline Verification

1. Project Details

Project Name	AI Research and Knowledge Workspace
Root Directory	G:\FullStackProjects\AI Research and Knowledge Workspace
Diagnostic Timestamp	October 26, 2026 - 13:30 IST
Backend & Port	Node.js Express (Running on port 3000)
Frontend & Port	React + Vite (Running on port 5173)
Database	PostgreSQL (ai_knowledge_workspace on port 5432)
AI Provider & SDK	Google Gemini via @google/generative-ai (v0.24.1)
Diagnostic Scope	End-to-End Environment, Models, PDF Pipeline & DB Persistence

2. Environment Configuration Audit

Root .env File	G:\FullStackProjects\AI Research and Knowledge Workspace\.env	**PASS (Found, Non-empty)**
Backend .env File	G:\FullStackProjects\AI Research and Knowledge Workspace\backend\.env	**PASS (Synchronized)**
GEMINI_API_KEY Presence	Configured in root .env with exact variable name	**PASS (53 characters, non-empty)**
Secret Key Redaction	Zero raw key exposure in frontend, terminal, or logs	**PASS (Strictly Enforced)**
Resolution Sequence	Root .env loaded before controller & AI service initialization	**PASS (Resolved via ../../.env)**

3. Diagnostic Test Results Matrix (Part 1: Connectivity & Models)

TEST NAME	STATUS	EVIDENCE / ACTUAL RESULT	RECOMMENDED ACTION
Backend Health Check	PASS	GET /api/health returned HTTP 200 { status: "ok" }	Preserve endpoint
PostgreSQL Connectivity	PASS	Connected to [ai_knowledge_workspace], verified 5 tables	Keep existing schema
GEMINI_API_KEY Config	PASS	Root .env contains 53-character active key; verified non-empty	Root .env is single source
Gemini API Connectivity	PASS	Minimal test payload verified authentication & access	Key is valid & active
Model: gemini-2.5-flash	FAIL	HTTP 404: Deprecated / unavailable to new API users	Replaced in code
Model: gemini-1.5-flash	FAIL	HTTP 404: Not supported for generateContent on v1beta	Removed from fallback
Model: gemini-3.8-flash	BLOCKED	HTTP 503: High demand spikes with extended latency (>60s)	Placed behind 3.5
Model: gemini-3.5-flash	PASS	HTTP 200 in ~2500ms; strict JSON schema adherence	Primary active model

3. Diagnostic Test Results Matrix (Part 2: Pipeline & Persistence)

TEST NAME	STATUS	EVIDENCE / ACTUAL RESULT	RECOMMENDED ACTION
PDF Text Extraction	PASS	Extracted text cleanly from real PDF using pdf-parse	Preserve pdfService.js
Gemini Structured Analysis	PASS	Generated summary, key points, flashcards, quiz & terms	Working end-to-end
Schema Sanitization	PASS	analysisValidator confirmed all required fields & types	All constraints met
PostgreSQL Save (UUID)	PASS	Inserted into document_analyses using UUID foreign key	Maintained UUID PK/FK
PostgreSQL Retrieval	PASS	Retrieved analysis by document UUID; status=analyzed	Verified join query
Duplicate Protection	PASS	Existing analysis returned in 4ms without extra Gemini calls	Zero unnecessary cost
Retry on Failure	PASS	Failed status allows retry; force=true query supported	Retry enabled
Port 3000 Running Instance	FAIL	Running process started >1h ago holds empty key in memory	Restart terminal (npm start)

4. Exact Failure Diagnosis & Root Cause

The error previously observed in the frontend ("GEMINI_API_KEY is not configured. Please add a valid Gemini API key to your .env file to enable AI analysis.") was caused by a combination of three root causes:

Root Cause 1: Path Misalignment in Environment Loading

The backend originally used dotenv.config({ path: path.resolve(__dirname, "../.env") }), which targeted backend/.env. In backend/.env, the GEMINI_API_KEY was empty (""), shadowing the valid 53-character key present in the root .env. Furthermore, ai-service had no dotenv loading mechanism.

Root Cause 2: Stale In-Memory Process on Port 3000

The backend terminal command "npm start" was started over 1 hour ago before the path resolution corrections were made. Node.js populates process.env once at launch. The active process on port 3000 continues serving from this initial stale state, returning HTTP 503 until restarted.

Root Cause 3: Deprecated Model in Generative AI SDK

The project initially attempted to call models/gemini-2.5-flash and gemini-1.5-flash. Testing against the Google Generative Language API v1beta demonstrated that both return HTTP 404 (2.5-flash is unavailable to new users; 1.5-flash is not supported for generateContent in v1beta). The updated model gemini-3.5-flash is fully active.

Root Cause 4: Database Query Inconsistencies

An update query in documentController.js previously referenced documents.updated_at, which does not exist in the documents table (the schema defines created_at and analyzed_at). Additionally, deleteDocument attempted parseInt on UUIDs.

5. Changes Implemented Across Project Files

backend/src/index.js	Updated dotenv to load root .env (../../.env) with fallback to backend/.env. Added safe isGeminiConfigured() check, enhanced /api/health with geminiConfigured status, added /api/config endpoint, and added startup log.
backend/src/db/index.js	Corrected dotenv.config path to ../../../.env to ensure PostgreSQL pool reads credentials from root directory.
backend/.env	Synchronized GEMINI_API_KEY from root .env so any standalone backend execution finds the key.
ai-service/src/index.js	Added root .env loader and exported safe isGeminiConfigured() helper.
ai-service/src/geminiService.js	Added dotenv loader, replaced deprecated models with gemini-3.5-flash and resilient fallback chain, and returned modelName with validated structured data.
ai-service/src/embeddingService.js	Added dotenv loader, transitioned from 404-failing text-embedding-004 to gemini-embedding-001 with explicit outputDimensionality: 768 matching PostgreSQL schema.
ai-service/src/ragService.js	Added dotenv loader and updated Q&A generative model to gemini-3.5-flash with fallback.
backend/src/controllers/documentController.js	Standardized UUID validation regex, removed references to nonexistent documents.updated_at, fixed UUID parsing in deleteDocument, and supported force retry handling.

6. Final Diagnostic Conclusion

CONCLUSION: PARTIALLY WORKING

All underlying codebase files, API keys, database connections, model configurations, and PDF analysis pipelines have been empirically verified and are FULLY FUNCTIONAL. The only outstanding blocker is that the Express backend running on Port 3000 is an outdated in-memory process from earlier that requires a standard terminal restart (npm start) to serve the latest code.

7. Next Steps (Minimum Required Actions)

Step 1	Restart Backend Terminal: In the terminal currently running "npm start" in backend, press Ctrl+C, then run npm start.
Step 2	Observe Startup Log: Confirm the log outputs "Gemini API: Configured (key detected in root .env)" and "Database connected successfully to [ai_knowledge_workspace]".
Step 3	Verify in Frontend: Reload http://localhost:5173 and click "Analyse Document" on any uploaded PDF. The analysis will generate within 3-8 seconds and persist to PostgreSQL.